Maximum Loan Capacity Estimator

AI101 Mini-Project | Team 67

https://github.com/Anurag0916/AI101_MIni_Project

Project Team

Anurag Beri Entry No. 2026AIB1018	Execution Lead
Nikita Entry No. 2026MCB1550	Documentation Lead
Kushagra Gupta Entry No. 2026EEB1370	Video Lead
Aashish Kumar Entry No. 2026EEB1312	Presentation Lead

Problem & Dataset Landscape

The Challenge

Current Manual Process

Slow manual cycles. Subjective assessments.

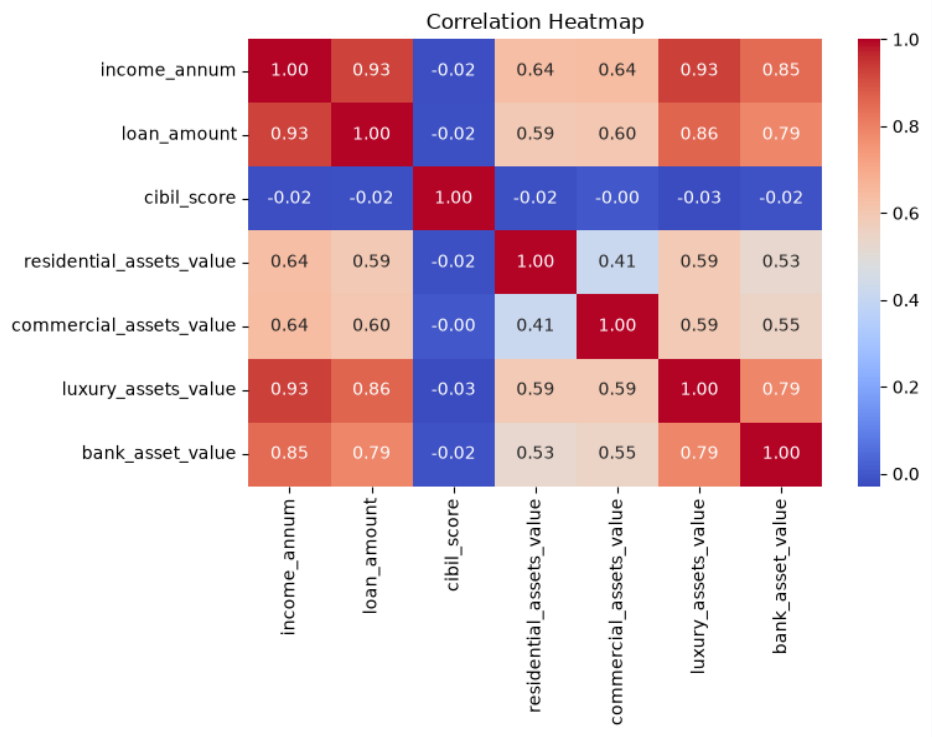
AI-Driven Solution

Instant, data-driven lending estimates.

Dataset Anatomy & Insights

4,269
TOTAL APPLICANTS

11
ANALYZED FEATURES



Approach and Pipeline

From raw record ingestion to model evaluation and inference.

1. Collect Data

4200+ applicants and 10 different features
Target:Max Loan Amount

2. Clean the Data

Removed duplicates, Filled missing values (median/most frequent) by imputer, Null values resolved

3. Preprocess

One-hot encoded the text-columns, Scaled numeric features,

4. Split and Train

80/20 train-test split, Linear Regression fitted on training set

5. Evaluate

R^2, MAE, RMSE on train and test. Residual Errors plotted.

WHY LINEAR REGRESSION?

Simple, highly interpretable coefficients allowing loan officers to audit exact weight assignments transparently.

LIBRARIES USED

Pandas · Numpy · Scikit-learn
Matplotlib · Seaborn

Model Evaluation

R2-SCORE

0.8514

Train:0.8626

MEAN ABSOLUTE ERROR

2626034.54

Train:2563779.09

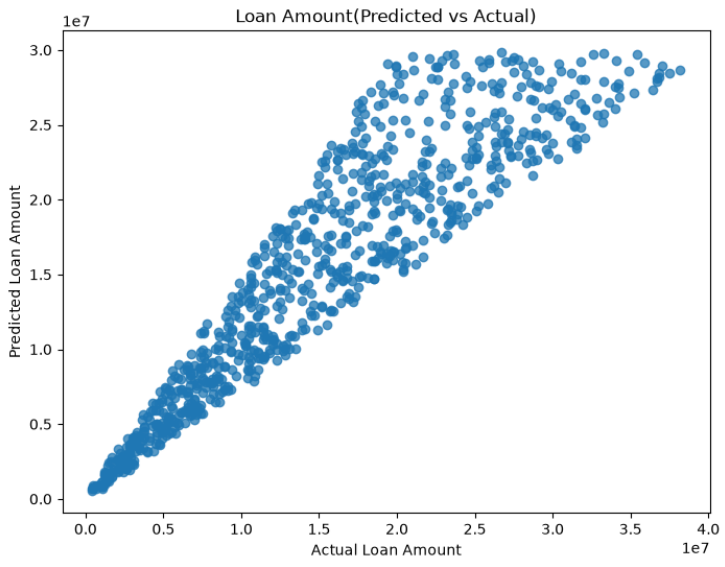
MEAN SQUARED ERROR

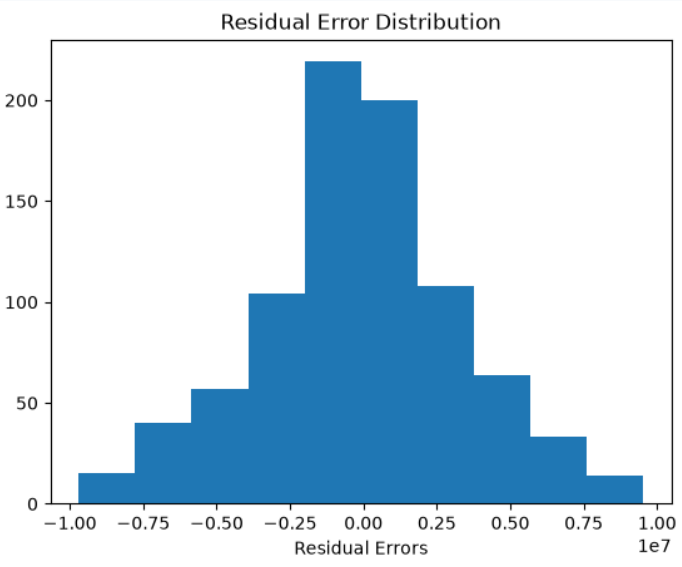
3451901.16

Train:3359832.40

Key Insights

- **High Variance:** Model captures financial patterns reliably.
- **Top Drivers:** Income tightly correlates with approval limits.
- **CIBIL Impact:** Credit scores heavily guide final predictions.





Learnings & Disclosures

Project Learnings

What Worked:

Linear Regression provided clear results. The plots show clean res

Challenges:

A file syncing error using git pull due to which some part of the code needed to be rewritten.
Error in the pipeline code which needed to be debugged.

Contributions

Anurag: Data loading, Model Training,

Nikita: Readme, Preprocessing

Aashish: Presentation, Preprocessing.

Kushagra: Evaluation Metrics, Video

AI-Use Disclosure

Claude and Gemini were used for debugging errors in code, create a heatmap and to solve a git pull conflict.

Gemini was used to generate the first draft of README. Edited By Nikita.
AI tools were used to create the template of the presentation. Edited by Aashish.